MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, PHILLIPS, RIBBED, CRUCIFORM, INSERT TYPE, EXTERNAL HEXAGON SHANK, MANUALLY DRIVEN

This specification sheet is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bits described herein shall consist of this specification sheet and the latest issue of MIL-B-85643.

1. REQUIREMENTS.

1.1 Material: UNS T41901 (AISI S-1) or UNS T41902 (AISI S-2) tool steel in accordance with QQ-T-570, Type S1 or S2.

1.2 Dimensions and configuration:

Bit and retaining groove: In accordance with Figure 1 and Table I.

Driver point: In accordance with Figure 1, Detail A, and Figure 2, Table II.

1.2.1 Cruciform driver bit gage: In accordance with Figure 3 and Table III.

1.3 Hardness: Rockwell 58 to 62 HRC.

1.4 Identification: Stamp military part number (see 1.8) and manufacturer's name, trademark or symbols.

1.5 Test recess: The test recess configuration and dimensions shall conform to Figure 4 and Table IV.

1.6 Minimum ultimate torque loads: In accordance with Table I and MIL-B-85643.

1.7 Torque fatigue loads: In accordance with Table I and MIL-B-85643.

1.8 Example of military part number:

M85643/1 - 2 - A

- A — Indicates larger "A" dimension (see Table I)
- 2 — Driver size number (see Table I)
- M85643/1 — Basic specification sheet number

AMSC N/A FSC 5120

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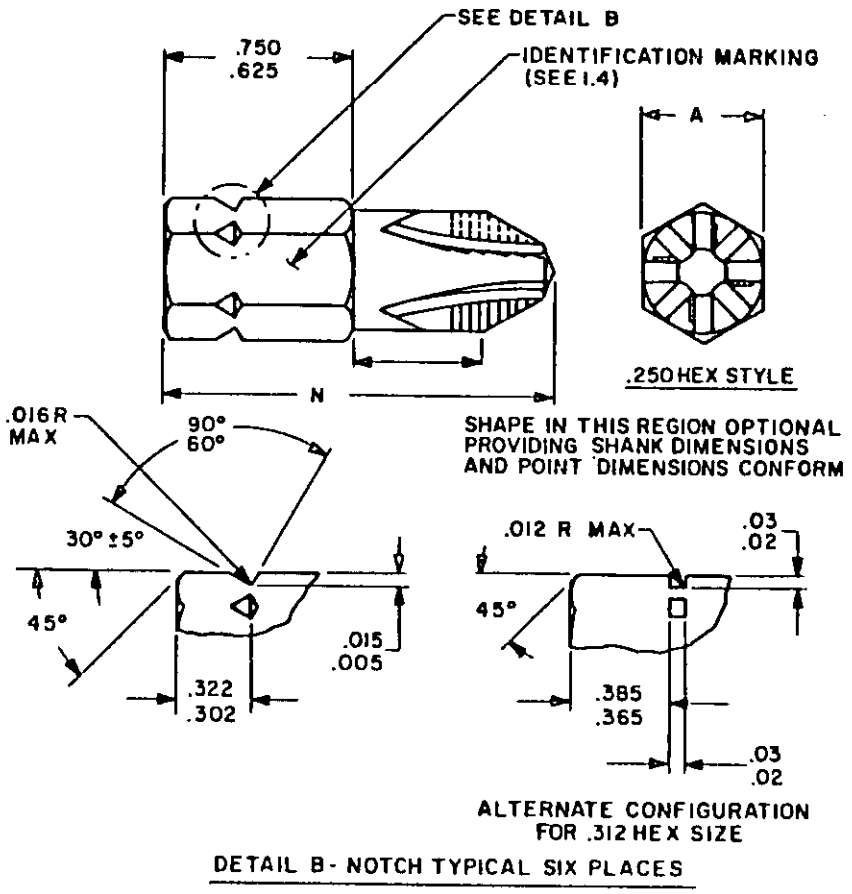


FIGURE 1. Bit.

TABLE I. Sizes and dimensions.

DRIVER SIZE NUMBER	RECESS SIZE	SCREW DIA SIZE RANGE FOR RECESS SIZE - REFERENCE	A +.000 -.004	W ±.12	MIN ULT TORQUE (LBF-IN.)	FATIGUE TORQUE (LBF-IN.)
02 02A	2	#6, #8, #10	.250 .312	1.00 1.25	90	45
03 03A	3	1/4	.250 .312	1.00 1.25	325	160

Dimensions are in inches.

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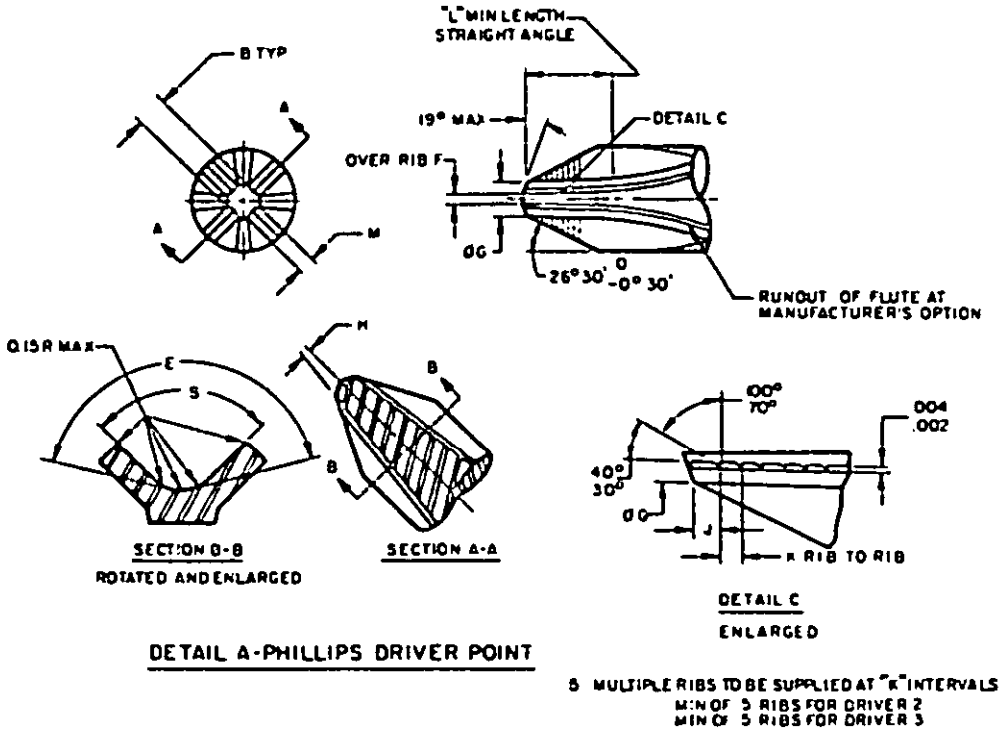


FIGURE 2. Driver point.

TABLE II. Phillips Driver Point Dimensions for ribbed cruciform configuration.

Driver Size	2	3
B ±.0010	.0606	.0983
E +0°30' 0	140°0'	146°0'
F Over Rib	.023-.025	.029-.031
G ±.001	.090	.150
H 0 -0°30'	5°45'	5°45'
J Max	.032	.045
K	.015-.022	.015-.022
L Min	.188	.281
M ±.001	.0434	.0826
S +0°30' 0	92°0'	92°0'

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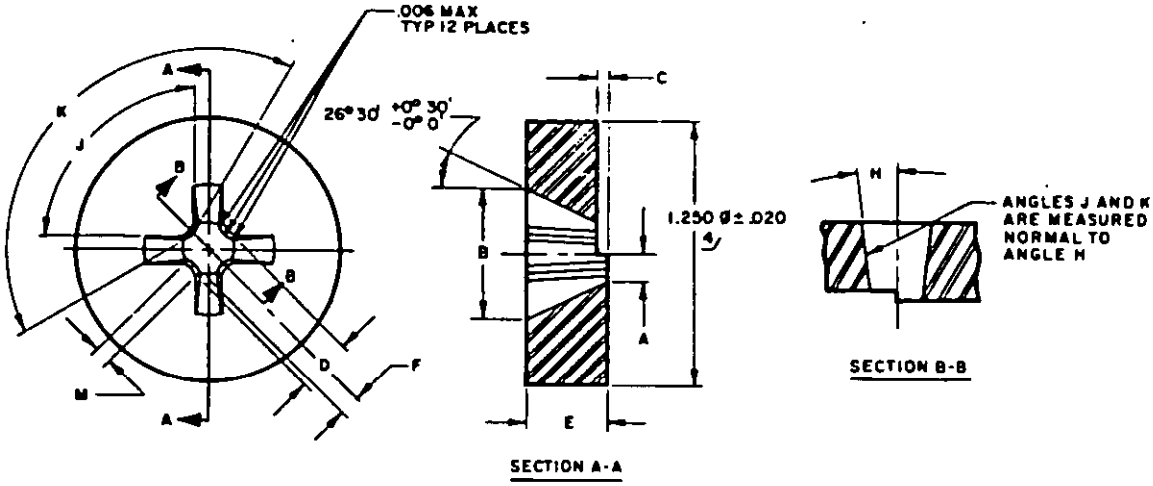


FIGURE 3. Driver point gage.

TABLE III. Dimensions for cruciform Phillips driver point gage.

Dash Number Driver Point Size	2	3
A ±.0002	.0455	.0755
B Ref	.278	.432
C ±.001	.010	.010
D Ref	.098	.156
E +.000 -.015	.188	.281
F +.0004 -.0000	.0303	.0496
H +0°15' -0°0'	5°45'	5°45'
J +0°0' -0°15'	92°0'	92°0'
K +0°0' -0°15'	140°0'	146°0'
M ±.001	.032	.079

All dimensions in inches.

NOTES:
1. Material: CRES shall conform to Class 440C per QQ-S-763 (UNS S44004).
2. Hardness: Rockwell 58-62 HRC.
3. Marking: Identify top surface with M85643, phrase "Driver Point Size" and applicable dash number indicating driver point size. Ex. M85643 - Driver Point Size 2.
4. The "G" dia of the driver point is to fall between the limits of "C" on this gage.

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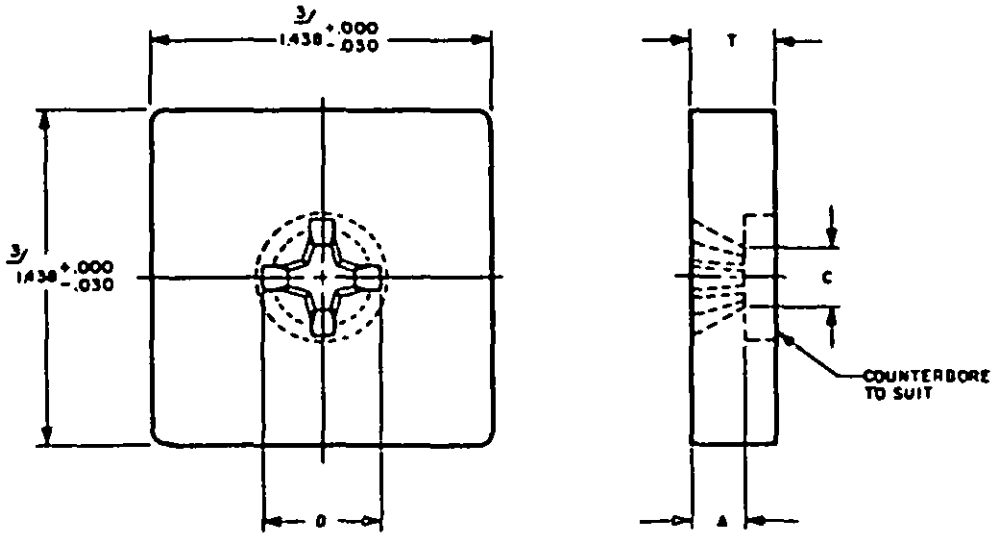


FIGURE 4.

TABLE IV. Dimensions for test recess for testing cruciform Phillips bits.

All dimensions in inches.

Recess Size	A Ref.	B ±.002	C ±.002	T ±.002
2	.143	.233	.090	.375
3	.236	.386	.150	

NOTES:
1. Material: Alloy steel (UNS T41902) type S2 per QQ-T-570, Type S-2, Type 6150, or equivalent.
2. Hardness: Rockwell 62-64 HRC.
3. Marking: Identify top surface with M85643 and recess size.
   Ex: M85643 - Recess Size 2.

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1.9 ACR and Phillips are registered trademarks of Phillips Screw Company. Patent 4,187,892 expires 12 February 1997. The Government does not have a royalty-free license.

Preparing Activity:
Navy - AS
(Project No. 5120-NB77)

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